



STANDING ORDER NO. HRD/54/2024

BASIC TRAINING OF ASI/SI - RADIO TECHNICIAN

I. INTRODUCTION

This standing order provides the guidelines for basic training of newly recruited ASI/SI - Radio Technician in Communication Unit of Delhi Police. The purpose of training is to prepare them professionally to discharge their duties according to their trade efficiently with requisite knowledge of Police Wireless Technical and also to inculcate in them, the habit of maintaining mental and physical health, discipline, punctuality and social skills. Besides, Outdoor activities such as Physical Training (PT), Drill, Parade etc., in Indoor activities, they will be imparted training wrt subjects of trade as well as knowledge of Police Organization, Rules & Regulations of Police Department & other such activities necessary for becoming a Police officer & to improve their knowledge of day-to-day Police wireless equipment's etc. in the interest of goals of Delhi Police.

II. OBJECTIVES OF TRAINING

The objective of their basic training is as follows: -

- (i) To inculcate force discipline.
- (ii) To improve physical strength/resilience.
- (iii) To give extensive theoretical knowledge and intensive practical training in their trade so that they learn about its requirements in the force and the manner of doing it in a disciplined organization.

III. DURATION

The period of basic training of ASI/SI - Radio Technician will be of **06 (six) months** prescribed in notified Recruitment Rules (RRs), as follows:-

S. No.	Training	Period
1.	Basic Training, DPA, JK	12 weeks
2.	Practical Training in District/Units, Radio Workshops of Communication unit	14 weeks

IV. JOINING PERIOD

- (i) All trainees of a batch will join the training on the same day. However, maximum relaxation of 02 working days can be given in

exceptional cases by Director/Joint Director, Delhi Police Academy. Relaxation of 04 days can be given with the approval of Special Commissioner of Police, Training, Delhi.

- (ii) The whole batch of the particular trade enlisted in a year should undergo training during that year and no one shall be exempted from the basic training.

V. TRAINING NEEDS

The purpose of training is to prepare them professionally to perform their duties according to their trade efficiently and effectively and also to inculcate in them the habit of maintaining mental and physical health, hygiene, discipline and punctuality, courtesy, straight-forwardness and requisite knowledge of office and Police procedures.

VI. TRAINING ASPECTS

The training at Delhi Police Academy shall be in two folds i.e. "**Outdoor**" aspect for all round physical fitness, capacity building, discipline and ability to perform vigorous duties in general; "**Indoor**" and Practical training for imparting professional knowledge, develop skills sets of the subject/Trade and dynamic changes in the field of applied knowledge.

Deputy Director, DPA will be the overall In-charge of Training at DPA, JK. The schedule for Basic Training, syllabus of Outdoor Training/examination, Indoor Training/examination is as at *Annexure - I, II & III* respectively.

A. Outdoor Training

The Outdoor training shall be supervised by the Asstt. Director, DPA who will be the In-charge Outdoor training assisted by the Chief Drill Instructor (CDI), DPA.

- (i) The training in outdoors shall be in accordance with the principle laid down in the Police Manual prescribed from time to time.
- (ii) The Chief Drill Instructor will issue weekly program after obtaining approval of Assistant Director, Outdoor.
- (iii) The daily duration/schedule of the outdoor training shall be as follows: -

Day time	Period	Activity	Duration
Morning	1	P.T.	40 Minutes
	2	Drill/Parade, Weapon Tactics & Training	40 Minutes (10 minutes break in-between)
	3	Yoga, Crowd Control, Traffic Control	40 Minutes
Evening	Games/Gym/UAC		60 Minutes

Details of Outdoor subjects, periods & marks are given at **Annexure -II**.

- (iv) In order to ensure effective training and proper supervision over the outdoor training, the trainees shall be divided into platoons of not more than 30 trainees, each under the command of a drill instructor as the platoon commander. He will be responsible for the maintenance of discipline and the general conduct of all the trainees of his squad. He will also ensure cleanliness of rooms and barracks under occupation by his trainees.
- (v) Team games will also form a regular part of outdoor training schedule.

B. Indoor Training

Indoor training shall be supervised by the ACP/Trg. /Comn, & AD/Indoor, DPA/JK assisted by the In-Charge Trg. /Comn. & CLI, DPA/JK.

- (i) The Indoor syllabus will cover General paper wrt Organization & Administration, Police & Society, Law etc. & the subjects of Police Wireless Technical Grade-II (Course Code: PWT-II) as approved by the **Directorate Coordination Police Wireless (DCPW) (MHA)**, as at Annexure - III.
- (ii) The training will be conducted according to the prescribed syllabus for which regular weekly time-table will be drawn.
- (iii) The main emphasis of the training will be on practical aspects of the trade. It will be the responsibility of DCP/Communication, Delhi to ensure that the Instructional Staff is well conversant with their subjects and are able to impart learning to the trainees. The selection of Instructional staff for teaching trade related aspects will be finalized by a committee headed by Joint CP/Operations.
- (iv) There will be 06 indoor periods (04 Theory and 02 Practical) and each period will be of 45 minutes duration & the indoor classes will run from 10:00 AM to 3:30 PM with a recess of 60 minutes duration.
- (v) The training courses will be conducted in English/Hindi medium for all candidates and they will have to learn and pass the test/examination in English/Hindi.
- (vi) Details of Indoor Subjects, Periods & Marks are given at **Annexure - III**.

VII. STUDY MATERIAL

The Joint CP/Operations, Delhi will constitute a committee under the Chairmanship of DCP/Communication in order to prepare comprehensive study material. The same shall be made available in the Delhi Police Academy library along with nominated faculty decided by the Committee. Digital content available on e-ustaad BPR&D, Knowledge Management System, i-Got Karamyogi & Delhi Police website/Nipun may also be used as training material.

VIII. GUIDANCE AND PROGRESS OF INDIVIDUAL TRAINEES

For effective supervision, the trainees shall be sub-divided amongst Inspectors/Sub Inspectors to be designated as "Course Guide" who shall be responsible for providing in-depth personal guidance to each trainee and the continuous supervision of progress in work, discipline and all-round conduct. A weekly interactive session will be conducted by the Course Guides to monitor the progress of the trainees. An Assessment of each trainee shall be done by the In-Charge/Trg. /Communication, Delhi & CLI, DPA/JK with the help of Course Guide to ensure improvement in professional skills.

Assistant Director/Training, DPA JK must ensure that the trainees are fully and comprehensively trained in all aspects. In all cases, where it is found that any trainee is lagging behind, special attention shall be paid so that she/he is brought to the desired level.

IX. ATTENDANCE AND LEAVES

- (i) 90% attendance in Outdoor/Indoor periods is compulsory, failing which she/he will be debarred from appearing in the examination. 10% relaxation in attendance may be given by Director/Joint Director, DPA in case of any unavoidable exigency.

The trainee who could not achieve the benchmark will be re-batched and be directed to proceed on leave without pay with the directions to join training from the beginning with the next scheduled batch. A trainee will be re-batched only twice and no further opportunity will be given thereafter & her/his services will be terminated by the competent authority under Rule 5 of the Central Civil Services (Temporary service) Rules, 1965.

Attendance of a trainee will be counted from the date of joining of training. However, those who join late, their syllabus will be covered by conducting extra classes.

- (ii) No leave should be granted during this period and only under exceptional circumstances it can be granted with prior approval of the Deputy Director.

X. DISCIPLINE

During basic training at Delhi Police Academy, the trainees shall be deemed to be under the administrative control of Director/Joint Director for all intent and purposes.

XI. EXAMINATION

The assessment of the trainee will be conducted in the form of Periodical Examination & Final Examination including the guidelines issued by DCPW (MHA) as under: -

A. Periodical Examination

Two internal assessment tests of the respective trade, involving theoretical & practical aspect related to their trades will be conducted after completion of every 01 month of training.

B. Final Examination

On completion of the basic training, they shall be examined by a Board of Examiners for Final examination, nominated by the Joint Commissioner of Police/Operations, Delhi on the request of DCP/Communication, Delhi. The board will be constituted under the overall supervision of DCP/Communication with 01 ACP from Communication unit & 01 Assistant Director from DPA, JK as members. Further, Director/Joint Director, DPA will nominate an Assistant Director, DPA and examiner/s (Inspector) for conducting Outdoor Examination.

- (i) In all there will be 03 papers for Indoor subjects and the examinees will be required to obtain marks as under, in order to pass the final examination: -

A.		Final Examination		Marks
1.	Paper - I	General Paper		100
2.	Paper - II	Theory (Trade)		100
3.	Paper - III	Practical (Trade)	Marks	100
		i. Measuring Instruments	25	
		ii. Fabrication & designing	25	
		iii. Project	25	
		iv. Presentation	25	
B.	Continuous Assessment: - Apart from final examination, performance of trainee will be assessed continuously on regular intervals by course instructors through unit tests etc.			Max Marks
1.	CA-I	Theory (Trade)		20
2.	CA-II	Practical (Trade)		20

Pass Criterion			Pass Marks
i	Paper - I	General Paper	50
ii	Paper - II	Theory (Trade)	40
iii	Paper - III	Practical (Trade)	40
iv	Aggregate Marks	80% of marks obtained in final examination + marks obtained in Continuous Assessment	

- (ii) The Board of examiner will consolidate the marks and declare the result. The training of those who do not pass would be extended by 02 (two) months and during this period, they would be given a maximum of 02 (two) additional chances in supplementary examinations to clear the subjects in which they have failed. However, only minimum qualifying marks i.e. 50% will be added in such subject(s) in the final result of the trainee, irrespective of the marks secured in such additional chances. If anyone does not pass

in all the papers even after availing the additional chances, his/her services will be terminated by the competent authority under Rule 5 of the Central Civil Services (Temporary service) Rules, 1965.

- (iii) After completion of their basic training successfully, they will report to Communication unit, Delhi Police for their practical training for a period of 14 weeks. 14 weeks training program will be chalked out by DCP, Communication for effective Hands-on training & skill development. On successful completion of their practical training, a certificate of proficiency will be issued by DCP/Communication which will be kept in their Service Book & updated in Intra DP also. A copy of the said Proficiency certificate must also be supplied to Training Branch, Delhi Police Academy, Jharoda Kalan, Delhi for record purposes.
- (iv) No ASI/SI - Radio Technician in Communication unit of Delhi Police shall be declared confirmed without passing their basic training & obtaining Proficiency Certificate in practical training from DCP/Communication.

XII. HONOUR LIST

The trainees will be eligible for the following awards for obtaining outstanding merit in various subjects at the occasion of the "Oath taking Ceremony" by Joint CP/Ops. or Director/Joint Director, Delhi Police Academy:-

1	All Round Best	Commendation Certificate with a cash Reward of Rs. 5,000/-
2	First in Indoor	Commendation Certificate with a cash Reward of Rs. 3,000/-
3	First in Outdoor	Commendation Certificate with a cash Reward of Rs. 3,000/-
4	First in Trade test	Commendation Certificate with a cash Reward of Rs. 3,000/-

In case the number of trainees is up to 15, then only 01 prize mentioned at S. No.1 will be given. If the number of trainees is above 15, but up to 50, then prize mentioned at S. No.1, 2 & 3 will be given. If the number of trainees is more than 50, then all the prizes mentioned above will be awarded. An entry to this effect will be made in the Service Book of the individual and updated in the Intra DP record also.

XIII. ATTESTATION PARADE

On the completion of course, an oath-taking ceremony will be held for the successful candidates. However, no trainees shall be sent to the Communication Unit or other District/Unit to which she/he is allocated for withdrawal of pay or for any other purpose whatsoever, unless she/he has successfully completed the training and has passed out of Delhi Police Academy.

XIV. CAUTION

While the trainee is undergoing the basic training course, Director/Jt. Director/Delhi Police Academy shall be the competent authority for initiating and finalizing disciplinary action against the trainee. He shall also be the competent authority to pass an order regarding termination of services of the trainee under The CCS (Temporary Services) Rules, 1965 & The Delhi Police (Punishment & Appeal) Rules, 1980. In order to eliminate any unsuitable trainee from the force at the stage of his/her training itself, any one of the following grounds shall be sufficient for termination of services: -

- a) Frequent absence of the trainee from Outdoor/Indoor course leading to a conclusion that she/he is an un-devoted trainee and a habitual absentee, not likely to render satisfactory services to the organization in future.
- b) Submission of false information and fake Medical Certificate from any doctor for availing leave or light duty or rest.
- c) A trainee who avails medical rest for a period of exceeding 14 days continuously will have to appear before a medical board for opinion, if required.
- d) Gross misconduct with the Indoor/Outdoor Instructors or with other trainees/staff.
- e) Any mischief or any crime involving her/his moral turpitude during the training period.
- f) If she/he commits any mischief or any crime involving her/his moral turpitude during her/his training.
- g) If any information submitted by the trainee, which is found to be false, fabricated or manipulated to gain employment.
- h) Any other ground which adversely reflects upon her/his behavior and the employee being found unsuitable for Police services.

XV. MEDICAL CONDITION/HEALTH STATUS & AGE FACTOR

- (i) Women trainee who is pregnant will not be allowed to join the training. Women trainees who are incapacitated on account of pregnancy during training will be re-batched. She will be allowed to join training from the beginning with the next scheduled batch within 2 years.
- (ii) Trainees suffering from serious ailments/diseases or who had sustained a physical injury resulting in a medical disability before joining the training/during training, such cases will be considered for appropriate remedy/facilitation i.e. writer/translator etc. with the approval of Deputy Director, DPA on the recommendation of the Assistant Director, Training.

XVI. DRESS CODE FOR TRAINING

The dress code for the trainees for P.T., Drill, Parade, Indoor Classes, Practical Training, Sports, etc. for summer and winter as well as for the attestation-cum-passing out parade shall be specified by the Director/Joint Director, Delhi Police Academy.

XVII. INTER-SE-SENIORITY

The Inter se-seniority shall be fixed as per the provision of Rule - 22 of Delhi Police (Appointment & Recruitment) Rules, 1980.

XVIII. REPRESENTATION

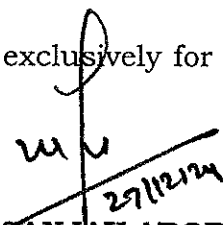
No representation with regard to result of any examination will be entertained after 30 days of the declaration of the result.

XIX. RELAXATIONS

The Commissioner of Police Delhi may relax any conditions laid down in this Standing Order as per the requirement from time to time.

XX. DISCLAIMER

It is made clear that this Standing Order is exclusively for internal smooth functioning of Delhi Police Department.


(SANJAY ARORA)
COMMISSIONER OF POLICE:
DELHI.

No. 16451-16600 /Record Branch/PHQ dated Delhi, the 27/12/2024

Copy forwarded to the: -

1. All Special Commissioners of Police, Delhi.
2. All Joint Commissioners of Police including Joint Director, Delhi Police Academy, Delhi/New Delhi.
3. All Additional Commissioners of Police, Delhi.
4. OSD to C.P., Delhi.
5. All Deputy Commissioners of Police, Districts/Units including PHQ, C.P. Sectt., FRRO and Deputy Director/Delhi Police Academy, Delhi.
6. DCP/HQ(IV)/PHQ with the direction to upload the Standing Order on Intra-DP.
7. LA to CP and FA to CP, Delhi.
8. All ACsP/PHQ/Insprs./PHQ, Delhi.
9. PS/Reader to CP, Delhi.
10. HAR/PHQ with 10 spare copies.
11. Inspr./Knowledge Center/PHQ.

SCHEDULE FOR BASIC TRAINING OF ASI/SI - RADIO TECHNICIAN**Distribution of Training Days**

S. No.	Particulars			Days
1.	Duration of the course (12 weeks - 12 x 7 = 84 days)			84
2.	Sundays, 2 nd Saturdays & Gazetted Holidays			20
3.	Number of days for Final Examination			03
4.	Total number of days available for training in DPA			84 - 23 = 61 days
5.	Total number of days for Outdoor Training at DPA			61
6.	Total periods available for Outdoor Training (61 x 4)			244 periods
7.	Total periods available for Indoor Training (61 x 6): -			366 periods
	General Paper	61 x 4 = 244 Periods	65 Periods	
	Theory (Trade)		179 Periods	
	Practical (Trade)	61 x 2 = 122 Periods	122 Periods	

Note: -

- A week will consist 06 (six) working days except 2nd Saturday/Gazetted Holidays.
- Indoor period will be of 45 minutes duration, while in Outdoor, PT will be of 40 minutes duration and Drill/Weapon Training/Crowd Control/Traffic Control/Yoga will also be of 40-40 minutes duration with 10 minutes break in Morning.
- Outdoor training in the evening (60 Minutes) will be for Games of choice/Gym/UAC but compulsory.
- In case of Law & Order duties, Gazetted Holiday/2nd Saturday/Sunday will be deemed as working days.
- However, this schedule is subject to change/modification in case of an exigency/contingency/administrative requirement by the Director/Joint Director, DPA.

OUTDOOR TRAINING SYLLABUS/EXAMINATION FOR
BASIC TRAINING OF ASI/SI - RADIO TECHNICIAN

S. No.	Subjects	Periods	Marks
1.	Physical Training (PT)	61	30
2.	Drill/Parade	50	30
3.	Weapon Tactics & Training	25	20
4.	Yoga	22	05
5.	Basics of Crowd Control	15	10
6.	Basics of Traffic Control	10	05
7.	Games/Gym/UAC (Evening)	61	--
TOTAL		244	100

Note: -

- CDI/DPA will provide syllabus/Test key for Outdoor Examination to the Board of Examiners duly approved by Director/Joint Director, Delhi Police Academy.
- Moreover, Outdoor training syllabus is subject to change/modification in case of an exigency/contingency/administrative requirement by the Director/Joint Director, DPA.

DETAILED OUTDOOR TRAINING SYLLABUS/EXAMINATION FOR
BASIC TRAINING OF ASI/SI - RADIO TECHNICIAN

S. No.	Subject	Period	Marks
1.	Physical Training (PT) - Physical fitness stamina building strengthening of arms, Pull-ups, Push-ups, sit-ups etc. - Ground work (Front, Back roll etc.) - Shuttle Race - Lifting and carrying	61	30
2.	Drill/Parade - Turnout, wearing of uniform - Attention, Stand at ease - Turning - Marching (Slow & Quick) - Salutes - Marching & Kadam Tall - Formations	50	30
3.	Weapon Tactics & Training Training of Short-range weapons will be as follows: - - Glock Pistol - 10 marks - MP 5 - 10 marks JVPC, 9MM Pistol, INSAS & AK-47 Rifle - <i>only for Familiarization</i>	25	20
4.	Yoga - Standing Asanas - Sitting Asanas - Lying position Asanas - Pranayam - Yog Nidra	22	05
5.	Crowd Control - Types of Crowds - Crowd Control with Rope and Barricades - Cane Shield Drill	15	10
6.	Traffic Control - Principle of Traffic Control - Road Sign and discipline - Road Signal - Manual control by hand drill, 2-3 & multiphase traffic control through hand drill - Classification of Vehicles	10	05
7.	Games of choice/Gym/UAC in Outdoor training in the evening	61	--
GRAND TOTAL		244	100

**INDOOR TRAINING SYLLABUS/EXAMINATION (THEORY & PRACTICAL) FOR
BASIC TRAINING OF ASI/SI - RADIO TECHNICIAN**

S.No.	Subject	Days x Periods		Total Marks
1.	Paper - I General Paper	61 x 4 = 244	65	100
2.	Paper - II Theory (Trade) as per SCR 2019 (Police Wireless Technician Grade - II)	Periods	179	100
3.	Paper - III Practical (Trade) as per SCR 2019 (Police Wireless Technician Grade - II)	61 x 2 = 122 Periods	122	100
TOTAL			366	300

PAPER - I

GENERAL PAPER

Organization, Administration, Police & Society, Laws & Service rules

Periods - 65**Marks - 100**

Time - 03 Hours

S.No.	Subject	Periods
1.	Organization & Administration • Organization of Central Government • Organization of State government, District & Sub-Division • Hierarchical set up of Delhi Police, Ranges, District, Sub-Divisions, Police Station, Various units of Delhi Police • Commissionerate System of Police • Badges/Flags (of cars) of all rank of police • Establishment of Communication unit & it's working • Personal hygiene, health and care • Turnout and care of Uniform • Discipline in police force - Rules & regulation etc. • Clothing & equipment • Pay and allowances, Income Tax Calculation • Rewards and decorations • Leave Procedure • Retirement benefits • Administration • Medical Benefits	20 02 02 03 01 01 01 01 01 01 01 01 01 01 01 01 01 01
2.	Police and Society • Organization of Political & Communal parties. • Understanding human behavior ○ Individual ○ Group behavior • Police public relations ○ It's Importance ○ Interpersonal relationship • Expectations of the people ○ Community Policing • Human Rights ○ Concept of Human rights and their importance in Police work ○ Do and Don'ts for protection and observance of human rights ○ Provisions of Human Rights in Indian Constitution & in other laws ○ Human Rights Organization & civil liberties ○ Uplift of weaker sections of the society ○ Gender Sensitization	15 02 03 04 02 04

3.	Laws & Rules – Basic Concepts • Constitution of India - Salient features • Fundamental rights, duties & directive principles • Articles 303 to 311 of the Constitution of India • General Structure of The Bharatiya Nyaya Sanhita, 2023 • General Structure of The Bharatiya Nagarik Suraksha, Sanhita, 2023 • General Structure of Delhi Police Act, 1978 • Delhi Police (Promotion & Confirmation) Rules, 1980:- Rules 4 to 19 & 23 • Delhi Police (General Conditions of Services) Rules, 1980:- Rules 7, 8, 10 to 13, 15, 16 & 17 • Delhi Police (appointment & Recruitment) Rules, 1980:- Rules 3 to 5, 7 to 11, 14, 21 to 25, 28 & 29 • CCS (Central Civil Services) Rules • CCA (Classification, Control & Appeal) Rules 1965	22 02 02 02 03 03 02 02 02 01 02 01
4.	Fire Fighting • Elementary knowledge of Fire-fighting • Equipment for Fire-fighting - Practical exercise.	03
5.	First Aid • General Principles of First aid and importance • Choking, its Causes and First aid • Application of bandages for control of bleeding • First Aid in situation of Fracture, Burn injuries, Snake-bite & electrocution cases • Basic Life Support Awareness/Cardio Pulmonary resuscitation (CPR)	05
TOTAL		65

PAPER – II
THEORY (TRADE)

Periods - 179

Marks – 100 (20+80)

Time - 3 hrs

UNIT - I	ENGINEERING MATHEMATICS
1.	Coordinate Geometry: Coordinates, Equation of straight line in various Standard forms, slope of line
2.	Algebra: Vector algebra, addition, subtraction unit vectors, angle between Two vectors
3.	Calculus concept of limit, rate of change, differentiation of simple algebraic and trigono-metrical functions: definite and indefinite integrals.
UNIT-II	ELECTRONIC DEVICES AND CIRCUITS
4.	Special Electronics Devices: Working principles and characteristics of Zener diode, Schottky diode, Varactor diode, Photo diode, Light emitting diode, SCR, Applications of TRIAC, DIAC, UJT.
5.	Power Supply: Regulated Power supply (circuit diagram), SMPS (block diagram), Safety devices in Power supplies, filter circuits in power supplies, Electronic-voltage regulator circuits, SCR applications, special cutoff. Circuits for low and over voltage.
6.	Batteries: Introduction to types and chemistry of batteries, Nickel Cadmium (Ni-Cd), Nickel-Metal Hydride (Ni-MH), Lead Acid, Lithium Ion (Li-ion), Lithium-ion Polymer, battery charging at various rate, maintenance of batteries.
UNIT-III	ELECTRONIC CIRCUITS AND ANALYSIS
7.	Field Effect Transistor: Construction-working principle of FET classification of FET-characteristics of FET-difference Between FET and BJT Applications of FET as amplifier, MOSFET and its types and applications
8.	Wave Shaping Circuits: A-stable, Mono-stable and Bi-stable Multi vibrators using Transistors, wave shapes-Schmitt Trigger using Transistors, IC-555 and its uses.
9.	Power Amplifiers: Difference between voltage & power amplifiers; importance of impedance match in power amplifier, class A, class B, class C & class AB Amplifier; working principles of push pull amplifier circuits, its advantages over single ended power amplifier
UNIT-IV	DIGITAL ELECTRONICS & CIRCUITS
10.	Combinational Circuits: Arithmetic circuits, Binary – Addition, subtraction, 1's and 2's complement Boolean algebra – Boolean theorems, minimization of Boolean function, K-Map, Basic logic-gates. Boolean functions realization using logic-gates, Half & Full adder, subtractions, Encoder and decoder - 3 to 8 decoder, BCD to seven segment decoders.

11.	Sequential Circuits: FLIP FLOPS – SR, JK, T, D, Counters: Asynchronous counters: up/down counter. Asynchronous and Synchronous counters, Ring counter Johnson counter with timing diagram. Basic functions and block diagram of MUX, DEMUX, Encoders and Decoders
12.	Operational Amplifier: Block diagram of an OPAMP, pin diagram of IC 741, Characteristics of an Ideal OP AMP - Simple Equivalent circuit of an OPAMP - virtual ground - OPAMP parameters - CMRR - Slew rate Applications of Operational amplifier as Inverting and Non-Inverting amplifier
UNIT-V	TRANSMISSION LINES
13.	Transmission Lines & Wave Guide: Line equation, characteristics impedance loss less line, Voltage standing wave ratio (VSWR), Transmission line as circuit element, Quarter wave transformer, Impedance matching, single stub, Wave between parallel planes, TE waves, TM waves, characteristics of TE and TM waves.
14.	ANTENNA & WAVE PROPAGATION: Definition-types of antenna: Mono pole and dipole antenna, directional and Omni directional antenna, Dipole arrays, Yagi antenna, Log periodic antenna, directive gain, radiation pattern, Radio wave propagation, Ground wave propagation, sky wave, space wave propagation, ionosphere layers
15.	Networking Component: Different types of networking cables, various connectors: shielded and unshielded cables, cross cable, straight cables, Various standards of Ethernet cable: 10 baseT, 100 baseT etc.
UNIT-VI	TELECOMMUNICATION TECHNIQUES
16.	Multiplexing: Frequency division Multiplexing(FDM), Time Division Multiplexing(TDM), Wavelength Division Multiplexing(WDM), comparison of FDM and TDM
17.	Wave form Coding Techniques: Sampling theorem. Pulse modulation, Pulse co-demodulation Differential PCMD delta-modulation and its drawbacks, Adaptive Delta-modulation.
18.	Types of Networks: LAN-MAN-WAN-Intranet-Extranet, Client-Server, Peer to Peer Networks. Topologies: Star, Bus, Ring, Mesh, Hybrid – Advantages and Disadvantages of each topology.
UNIT-VII	COMMUNICATION SYSTEMS
19.	Cellular System: Frequency reuse, channel assignment strategies, handoff strategies, interference and system capacity, fading & multipath propagation, types of fading, Telecommunication systems, GSM, DECT, TETRA, UMTS & IMT-2000, 2G, 3G, 4G, LTE, LTE advanced.
20.	Optical Fiber Communication Systems: Block Diagram of, Optical fiber communication system, Principles of Light Transmission in a Fiber, critical angle, acceptance angle, acceptance cone, Single Mode,

	Multimode Fibers, Step Index, Graded Index, Losses in optical fiber, Optical Sources – LED – Semiconductor LASER, Optical Detectors – PIN and APD
21.	Satellite Communication System: Basics of satellite, Satellite orbits, Geo- stationary orbits, satellite components, transponder, frequency bands, Ground station, block diagram of satellite communication system, EIRP, antenna gain, G/T, VSAT communication
UNIT-VIII	POLICE COMMUNICATION SYTEMS
22.	HF Transceivers: Detailed study of HF transceivers with the help of circuit Diagram like MHM300HFTX/RX. Measurements and trouble-shooting of HF radios.
23.	VHF Transceivers: Detailed study of VHF transceivers with the help of circuit diagram like GP300, VHFTX/RX. Measurements and trouble-shooting of VHF radios.
24.	POLNET: Detailed study of functioning of POLNET Network, installation of POLNETVSATs, maintenance, measurements applications & trouble Shooting of POLNET Terminals.
25.	Telecom Regulations: Telegraph Act, Wireless Act, various Licenses and their procedure (SACFA/NOCC/decision letter/AIP) etc. Importance Cyber Security in police communication, various Act related to cyber security, IT Act.
Technological Update	
26.	NVIS Technology and its importance in HF communication, Half - Loop Antenna

PAPER – III
TRADE (PRACTICAL)

Periods - 122

Marks - 100 (20+80)

A)	FABRICATION & DESIGNING
1.	To design & fabricate Two Stage RC Couple Amplifier
2.	To design & fabricate RC Phase Shift Oscillator
3.	To design & fabricate Mono-stable, Bi-stable and A-stable Multi vibrator circuits
4.	To design & fabricate Push-Pull Amplifier
5.	To design & fabricate Regulated Power supply
6.	Logic gates fabrication over multipurpose PCB by soldering iron
7.	To study working of D, JK, and T flip flop using digital trainer board
8.	To study working of MUX and DEMUX with help of Digital Trainer Board
B)	STUDY OF INSTRUMENT, TRANSCEIVERS & INSTALLATION OF NETWORK
9.	Detail study of 15wHF TX/RX, specification, measurement
10.	Detail study of VHF TX/RX, specification, measurement
11.	Study of Optical fiber communication
12.	To study and observe Passive filters over trainer board.
13.	Field Exposure: - Installation & Establishment of VSAT Communication
C)	COMPUTER FUNDAMENTALS
14.	Computer Networking (LAN Installation), File transfer
